

Business Problem Statement – Swire Coca-Cola

TEAM 5 - IS 6813-001 Fall 2024

Business Problem Statement

Swire Coca-Cola has six production plants to meet the demand of 192 million cases of various beverages from 13 different states. These plants are currently only able to meet 94.4% of the demand. One of the reasons these plants fall short of meeting the total product demand is that they are plagued with unplanned downtime. This downtime is caused by various failure modes like maintenance, mechanical failure, general wear and tear, etc.

Extensive repairs result in lengthy downtimes that generate work orders in the IWC system. The long lead times caused by these work orders cost Swire Coca-Cola \$60 million annually in production losses.

The purpose of this project is to create a predictive model to identify machines at risk of failure to prevent downtime and optimize Swire Coca-Cola's plant machine maintenance.

Benefits of a Solution

Swire Coca-Cola needs to better understand these failures to plan predictive maintenance and procure critical parts before failure. This would allow them to reduce costly lead times and replace consumable parts outside of production windows. A few of the benefits of increased understanding are

- Understanding and quantifying the life cycle of machine parts
- Optimizing scheduling to maximize production output
- Develop order point quantities for critical machine parts and consumables.

Analytics Approach

Given the amount of time series data provided by Swire Coca-Cola our analytics approach will be a combination of time series analysis, classification models, and regression techniques. Before we deploy our models, the data will need to be preprocessed to feed our model normalized and clean data. From there, we will use classification models to describe and highlight relationships between variables. For the estimation of downtimes, we will use regression techniques. Prescriptive analytics can be used to set things like order point inventories for stocking essential machine parts.

Success Metrics

- The project will be considered a success if the model developed can describe some of the causes and effects and predict machine failures better than a random chance of 50/50.

Scope/Deliverables

Primary deliverables for this project will be the following:

- Predictive model capable of predicting machine part failure.
- A detailed report outlining model findings and performance.

- GitHub repository of all Python and R code.

Project Details

The members of Team 5 will execute this project. All deliverables will be provided to the Swire Coca-Cola team and MSBA faculty. The project is expected to be completed by November 26th, 2024, with the following milestones:

- Business problem statement delivery - September 15th.
- Exploratory data analysis - October 6th.
- Model development and testing - November 3rd.
- Final model delivery and evaluation report through a presentation and write-up - November 26th.